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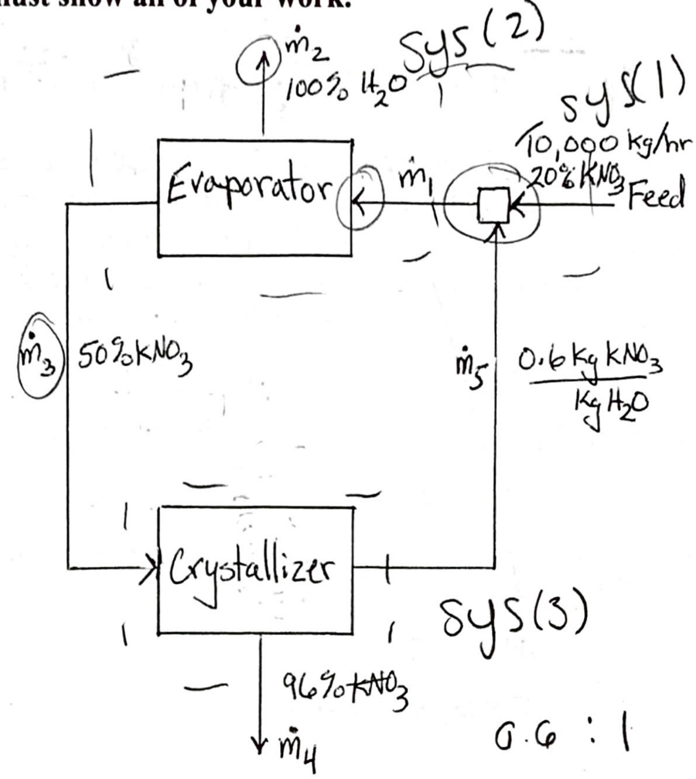
ChE 210 Exam #1 21 February 2020

To receive full credit, you must show all of your work.

(1) (25 pts.) Potassium nitrate (KNO_3) is separated as suspended solids from water (H_2O) by evaporation and then crystallization. A steady, continuous process is shown in the diagram. Calculate the five unknown mass flow rates $\dot{m}_1$, $\dot{m}_2$, $\dot{m}_3$, $\dot{m}_4$, and $\dot{m}_5$ in kg/hr. The feed enters at $\dot{m}_F$ = 10,000 kg/hr and 20 wt% KNO_3. Pure water leaves the top of the evaporator and a second stream of 50 wt% KNO_3 solution also leaves the evaporator. The crystallizer makes crystals that contain only 4 wt% water and a second saturated stream with 0.6 kg KNO_3/kg H_2O which is recycled with the feed. Clearly indicate where you define your system boundaries for any balances you write. These systems should be carefully chosen to make the problem easier to solve.



0.6 : 1

$\dot{m}_F = 10{,}000 \frac{kg}{hr}$ In = Out

20% KNO_3

1

$\dot{m}_F + \dot{m}_5 = \dot{m}_1$

$\dot{m}_1 = 10{,}000 \frac{kg}{hr} (.20\ KNO_3) + \dot{m}_5 \left(\frac{0.6 \text{ kg } KNO_3}{1 \text{ kg } H_2O}\right)$ $\boxed{\dot{m}_1 = 2000 + .6\ \dot{m}_5}$

$w_{1\,KNO_3}$?

3 $\dot{m}_3 = \dot{m}_4 + \dot{m}_5$

$\dot{m}_3 = \dot{m}_4(0.96) + \dot{m}_5 \left(\frac{.6 \text{ kg } KNO_3}{1 \text{ kg } H_2O}\right)$

2

$\dot{m}_1 = \dot{m}_2 + \dot{m}_3$

$10{,}000 \frac{kg}{hr} = \dot{m}_2 (X_{H_2O}) + \dot{m}_3 (.50\ KNO_3)$

$\dot{m}_2$

$\dot{m}_2 + \dot{m}_3 = 10{,}000 \frac{kg}{hr} (.20\ KNO_3) + \dot{m}_5 \left(\frac{0.6 \text{ kg } KNO_3}{1 \text{ kg } H_2O}\right)$ ok

$\dot{m}_2(1) + \dot{m}_3(.50\ KNO_3) =$

$\dot{m}_2 + \dot{m}_4(0.96) + \dot{m}_5 \left(\frac{\cancel{.6} \text{ kg } KNO_3}{1 \text{ kg } H_2O}\right) = 10{,}000 \frac{kg}{hr} (.20\ KNO_3) + \dot{m}_5 \left(\frac{\cancel{0.6} \text{ kg } KNO_3}{1 \text{ kg } H_2O}\right)$

$\dot{m}_2 + \dot{m}_4 (0.96) =$

1	7
2	15
3	28
T	50

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$\dot{m}_1 =$ ______________ kg/hr, $\dot{m}_2 =$ ______________ kg/hr, $\dot{m}_3 =$ ______________ kg/hr,

$\dot{m}_4 =$ ______________ kg/hr, $\dot{m}_5 =$ ______________ kg/hr

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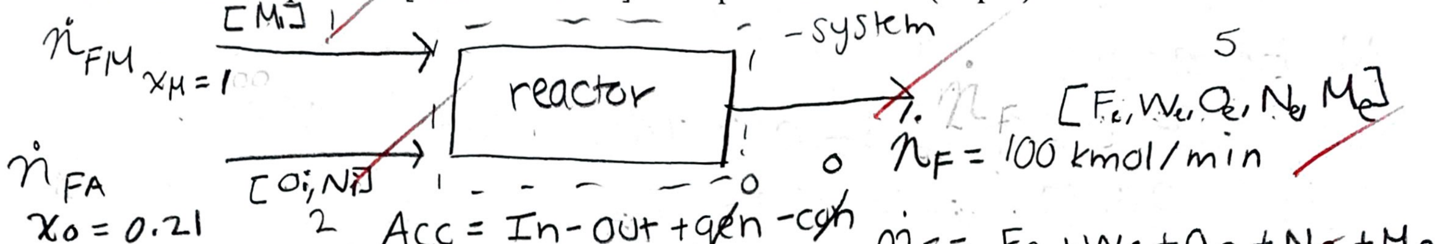
(2) (45 pts.) Formaldehyde (HCOH) is produced by oxidation of methanol (CH_3OH). A reactor is fed a stream of pure methanol $\dot{n}_{FM}$ and a stream of air (21 mol% O_2 and 79 mol% N_2) $\dot{n}_{FA}$. A single product stream of 100 kmol/min leaves the reactor containing formaldehyde, water (H_2O), oxygen, nitrogen, and methanol. Chemical analysis determined that the product stream is 3.4 mol% oxygen and 42.7 mol% nitrogen.

(a) Draw and label a block flow diagram of the process described above. Represent the species molar flow rates as N_i, O_i, and M_i for inlet nitrogen, oxygen, and methanol and N_e, O_e, M_e, W_e, and F_e for the exit nitrogen, oxygen, methanol, water, and formaldehyde. Notice that $\dot{n}_{FM}= M_i$ and $\dot{n}_{FA}= N_i + O_i$. Write a balanced chemical reaction equation for the process. What can you conclude about the mole fractions of the two reaction products [H_2O and HCOH] in the product stream? (10 pts.)



Acc = In - out + gen - con

In = out

$\dot{n}_{FM} + \dot{n}_{FA} = \dot{n}_F$

$\dot{n}_F = Fe + We + Oe + Ne + Me$

$x_{Oe} = 0.034$

$x_{Ne} = 0.427$

$\dot{n}_{FM} = Mi$

$\dot{n}_{FA} = Ni + Oi$

$Mi + Ni + Oi = Fe + We + Oe + Ne + Me$

$(Mi - Me) + (Ni - Ne) + (Oi - Oe) = Fe + We$

The mole fraction of F and W are equal to the total mole fractions from N, O, M.

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(b) Do a degree of freedom analysis following the method on page 105 of the textbook by Murphy. Fill in the table below giving a brief explanation for your entries. Compute the DOF. (5 pts.)

	Number	Explanation
4a. Stream Variables	1+2+5 = 8	Mi + Oi + Ni + Fe + We + Oe + Ne + Me = 8 components
4b. System Variable	1	reaction occurs in reactor ↳ oxidation CH_3OH
Total Independent Variables	9	8 + 1
5a. Specified Flows	1	$\dot{n}_F = 100$ kmol/min
5b. Specified Stream Compositions	2	N - 1 → 2 - 1 = 1 2 - 1 = 1
5c. Specified System Performances	0	
5d. Material Balance Equations	6	M, N, O, F, W, ~~Total~~
Total Independent Equations	9	
DOF	0	9 - 9 = 0 → specified

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(c) Using atomic balances, calculate the molar flow rates for the two feed streams in kmol/min for the methanol $\dot{n}_{FM}$ and the $\dot{n}_{FA}$. Also calculate the mole fractions of formaldehyde y_F, methanol y_M, and water y_W in the product stream. (30 pts.)

(3) (30 p

$\dot{n}_F = 100$ kmol/min

$\dot{n}_{FM} = M_i = 1$ $\qquad M_i = N_i + O_i$

$\dot{n}_{FA} = N_i + O_i = 1$

$\dot{n}_F = \dot{n}_{FM} + \dot{n}_{FA} \rightarrow 100 \frac{kmol}{min} = M_i + N_i + O_i$

No balance:

$0 = 0.79\,\dot{n}_{FA} - 0.427\,\dot{n}_F + gen^{0} - con^{0}$

$0.427\,\dot{n}_F = 0.79\,\dot{n}_{FA}$

$0.427\left(100 \frac{kmol}{min}\right) = 0.79\,\dot{n}_{FA}$

$\dot{n}_{FA} = 54.05 \frac{kmol}{min}$

$\dot{n}_F = \dot{n}_{FA} + \dot{n}_{FM}$

$100 \frac{kmol}{min} - 54.05 \frac{kmol}{min} = \dot{n}_{FM}$

$\dot{n}_{FM} = 45.95$

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$\dot{n}_{FM}$ = 45.95 kmol CH_3OH /min, $\dot{n}_{FA}$ = 54.05 kmol air /min, y_F = ________ mol HCOC/mol,

y_M = ________ mol CH_3OH /mol, y_W = ________ mol H2O /mol

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(3) (30 pts.) A new synthesis mechanism for dimethyl carbonate (DMC, $C_3H_6O_3$) is as follows:

Methane (CH_4) and water (H_2O) react to form carbon monoxide (CO) and hydrogen (H_2):

$$3\,CH_4 + 3H_2O \rightarrow 3CO + 3H_2 \quad \text{(R1)}$$

C: 1 H: 6 : O C: 1 O: 1 : H6

CO and H_2 react to form methanol (CH_3OH):

C: 2 H: 12 : O: 2

$$2CO + 4H_2 \rightarrow 2CH_3OH \quad \text{(R2)}$$

C: 2 O: 2 H: 12

CH_3OH, CO, and oxygen (O_2) react to form DMC and water.

$$2CH_3OH + CO + \frac{1}{2}O_2 \rightarrow C_3H_6O_3 + H_2O \quad \text{(R3)}$$

(a) Use a generation-consumption analysis to synthesize a reaction pathway to DMC based on the three reaction above with no net generation or consumption of CO or CH_3OH. Hydrogen is allowed as a byproduct. Show your calculations and write the net chemical equation for the process. (20 pts.)

	R1	R2	R3	Net Rxn
CH_4	-3	0	0	-3
H_2O	-3	0	+1	-2
H_2	+9	-4	0	5
CO	+3	-2	-1	0
CH_3OH	0	+2	-2	0
O_2	0	0	-1/2	-1/2
$C_3H_6O_3$	0	0	+1	+1

$3CH_4 + 2H_2O + \frac{1}{2}O_2 \rightarrow C_3H_6O_3 + 5H_2$

H: 16 H: 16

C: 3 C: 3

O: 2+1=3 O: 3

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NET RXN: $3\,CH_4 + 2H_2O + \frac{1}{2}O_2 \rightarrow C_3H_6O_3 + 5H_2$

(b) Calculate the fractional economy of the overall reaction. (Assume the atomic weights of carbon, hydrogen, and oxygen are: $M_C = 12$ g/mole, $M_H = 1$ g/mole, and $M_O = 16$ g/mole, respectively.) (10 pts.)

Compound	v_i	$M_i v_i$	M_i g/g·mol	
CH_4	−3	−48	16	−48
H_2O	−2	−36	18	−32
H_2	+5	+10	2	80
CO	0	0	0	0
CH_3OH	0	0	0	0
O_2	−½	−16	32	−8
$C_3H_6O_3$	+1	+90	90	
		0	158	

$$\frac{V_p M_p}{-158} = \frac{90}{158} = \frac{90}{158}$$

$= 0.569$

?

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Fractional atom economy = ~~0.57~~

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